

Shubh Srivastava

+91-8630414010 | shubh230405@gmail.com | GitHub | LinkedIn

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT)

B.Tech - Computer Science and Engineering

2024 – Present

CGPA: 8.59 (Current)

PROFESSIONAL EXPERIENCE

Vault Stack AI

Generative AI Intern

(Jun 2026 – Jul 2026)

- Explored and implemented the core mechanics of Retrieval-Augmented Generation (RAG), mastering the data flow from document ingestion to context-aware LLM generation.
- Researched the application of ontological frameworks within RAG pipelines to map complex data relationships and enhance semantic retrieval accuracy.
- Experimented with vector embedding generation and document chunking strategies to optimize query relevance and reduce model hallucinations.
- Gained hands-on experience with domain-specific Generative AI architectures, evaluating different approaches to integrating external knowledge bases with language models.

KEY PROJECTS

ASTRAM – Congestion Forecasting | LightGBM, XGBoost, FastAPI, Next.js

Flipkart Gridlock Hackathon

GitHub | Demo

- Engineered an end-to-end machine learning platform to forecast traffic congestion severity, incident priority, and road closure risks across Bengaluru.
- Architected a multi-model pipeline utilizing LightGBM with SMOTE over-sampling and an XGBoost classifier tuned for minority class recall.
- Implemented an online learning infrastructure featuring Population Stability Index (PSI) drift detection, automated incremental retraining, and Exponential Moving Average lookup updates.
- Deployed a FastAPI REST API and a Next.js/Streamlit interface integrating a custom rule-based Natural Language Understanding (NLU) chatbot to streamline field resource dispatch protocols.

RBI Regulatory Assistant | RAG, Qdrant, BM25, SQLite, FinBERT, DSPy

Generative AI

GitHub

- Engineered a production-grade AI search application for regulatory documents using a robust Retrieval-Augmented Generation (RAG) architecture.
- Architected a hybrid database system combining Qdrant for dense semantic vectors, BM25 for sparse keyword matching, and SQLite for relational facts and in-memory ontology graphs.
- Optimized query retrieval through Reciprocal Rank Fusion (RRF) and cross-encoder reranking, integrating FinBERT to apply a financial-domain semantic relevance boost.
- Implemented a strict Dual-Check Verifier and adversarial self-critique mechanism to validate numeric claims against structured data and ground responses in source text.

Rabuste Coffee Platform | MERN Stack, Next.js, Razorpay, Gemini API

GWOC 2026

GitHub | Demo

- Developed a comprehensive MERN stack web application for a specialty coffee café, featuring a responsive React frontend and a robust Node.js/Express backend.
- Engineered a digital ordering system supporting QR-based self-service, counter ordering, and scheduled pre-orders, integrated securely with the Razorpay payment gateway.
- Integrated the Google Gemini API to power an intelligent coffee discovery engine, a domain-restricted customer support chatbot, and an advanced admin analytics dashboard with predictive forecasting.
- Built a secure, JWT-protected admin panel to manage inventory, art galleries, workshop registrations, and automated consent-based customer email marketing campaigns.

TECHNICAL SKILLS

Languages:	C++, Python, JavaScript, TypeScript, HTML/CSS, SQL
Frameworks & Libraries:	React.js, Next.js, Node.js, Express.js, FastAPI, TensorFlow, Keras, LightGBM, XGBoost
AI/ML & Data Science:	Retrieval-Augmented Generation (RAG), DSPy, Pandas, NumPy, Scikit-learn
Databases:	MongoDB, PostgreSQL, MySQL, SQLite
Tools & Cloud:	Git/GitHub, Postman, Docker, Google Cloud

ACHIEVEMENTS

Flipkart Gridlock Hackathon 2.0 Qualified for the competitive national-level hackathon (May 2026 – Jun 2026)

Competitive Programming Achieved **Specialist** (Rating: 1485) on **Codeforces** and **1788** rating on **LeetCode**

Problem Solving Solved **600+** Data Structures and Algorithms (DSA) problems across various platforms

JEE Main 2024 Secured **99.56** Percentile (AIR: 7001) in the national engineering entrance exam (2024)